Homework 1 - Angelina Mascarini Serra - 13686003 6.7/10

(1) With $J_{\mu\nu}$ as the generators of the Lorentz group. 0.8/2

(a) Show that $i[J^{\alpha\beta}, J^{\gamma\sigma}] = g^{\sigma\beta}J^{\gamma\alpha} + g^{\gamma\alpha}J^{\sigma\beta} + g^{\gamma\beta}J^{\alpha\sigma} + g^{\alpha\sigma}J^{\beta\gamma}$

writing the infinitesimal Lorentz transformation as

$$\text{(I)} \quad \Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu$$

and knowing that any element of a Lie group can be written as

$$\Lambda^\mu{}_\nu = e^{\alpha_i J_i} = e^{\alpha^{\rho\sigma} J_{\rho\sigma}}$$

there must be a i here if α is real

(for 2 indices)

with J_i being the generators of the group

for the infinitesimal transformation let's Taylor expand it

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \alpha^{\rho\sigma}(J_{\rho\sigma})^\mu{}_\nu + \mathcal{O}([\alpha^{\rho\sigma}]^2)$$

Now using (I) $\quad \omega^\mu{}_\nu = \alpha^{\rho\sigma}(J_{\rho\sigma})^\mu{}_\nu$

other important fact is that $\omega_{\rho\sigma}$ is antisymmetric

so $\omega^\mu{}_\nu$ has 6 independent components, because of it it's interesting to right

$$\boxed{\omega^\mu{}_\nu = \frac{i}{2}\omega^{\rho\sigma}(J_{\rho\sigma})^\mu{}_\nu} \quad \text{(II)}$$

so $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \frac{i}{2}\omega^{\rho\sigma}(J_{\rho\sigma})^\mu{}_\nu$

as will be proved on question (4) $\omega_{\mu\nu}$ is antisymmetric

so $\omega_{\mu\nu} = -\omega_{\mu\nu}$ → $\omega_{\mu\nu} = \frac{1}{2}(\omega_{\mu\nu} - \omega_{\nu\mu})$ (III)

and let's use that $\omega_{\mu\nu} = \delta^\rho_\mu \delta^\sigma_\nu \omega_{\rho\sigma}$ (IV) (V)

so combining (III) and (IV) we obtain $\boxed{\omega_{\mu\nu} = \frac{1}{2}(\delta^\rho_\mu\delta^\sigma_\nu - \delta^\rho_\nu\delta^\sigma_\mu)\,\omega_{\rho\sigma}}$

using (II) and (IV) $\frac{i}{2}\omega^{\rho\sigma}(J_{\rho\sigma})^\mu{}_\nu = \frac{1}{2}(\delta^\rho_\mu\delta^\sigma_\nu - \delta^\rho_\nu\delta^\sigma_\mu)\,\omega_{\rho\sigma}$

$= \frac{\;}{2}(\delta^\rho_\mu g_{\sigma\nu} - \delta^\rho_\sigma g_{\nu\rho})\,\omega_{\rho\sigma}$

wrong sign, unless you define the transformation

→ $\boxed{(J_{\rho\sigma})^\mu{}_\nu = -i(\delta^\rho_\mu g_{\sigma\nu} - \delta^\rho_\sigma g_{\nu\rho})}$ (VI) as $e^{-i\omega\frac{J}{2}}$ which is equivalent to switch $J \to -J$

carefull with the index placement. It does make difference.

for $[J_{\alpha\beta}, J_{\rho\sigma}]^\mu{}_\nu = (J_{\alpha\beta})^\mu{}_\nu (J_{\rho\sigma})^\lambda{}_\nu - (J_{\rho\sigma})^\mu{}_\lambda (J_{\alpha\beta})^\lambda{}_\mu$

so $(J_{\alpha\beta})^\mu{}_\lambda (J_{\rho\sigma})^\lambda{}_\mu = -(\delta^\mu_\alpha g_{\beta\lambda} - \delta^\mu_\beta g_{\alpha\lambda})(\delta^\lambda_\rho g_{\sigma\nu} - \delta^\lambda_\sigma g_{\rho\nu})$

reorganizing

→ $(J_{\alpha\beta}J_{\rho\sigma})^\mu{}_\nu = -[\delta^\mu_\alpha g_{\beta\rho}g_{\sigma\nu} - \delta^\mu_\alpha g_{\beta\sigma}g_{\rho\nu} - \delta^\lambda_\beta g_{\rho\alpha} + \delta^\mu_\beta g_{\alpha\sigma}g_{\rho\nu}]$

so with $[J_{\alpha\beta}, J_{\rho\sigma}]^\mu{}_\nu = (J_{\alpha\beta}J_{\rho\sigma})^\mu{}_\nu - (J_{\rho\sigma}J_{\alpha\beta})^\mu{}_\nu$

+ signs?

$= -[\delta^\mu_\alpha g_{\beta\rho}g_{\sigma\nu} - \delta^\mu_\alpha g_{\beta\sigma}g_{\rho\nu} - \delta^\lambda_\beta g_{\alpha\rho}g_{\sigma\nu}\,\delta^\mu_\beta g_{\alpha\sigma}g_{\rho\nu}]$
$+ [\delta^\mu_\rho g_{\sigma\alpha}g_{\beta\nu} - \delta^\mu_\rho g_{\sigma\beta}g_{\alpha\nu} - \delta^\mu_\sigma g_{\rho\alpha}g_{\beta\nu} + \delta^\mu_\sigma g_{\rho\beta}g_{\alpha\nu}]$

using (IV) it's possible to recognize the generators so unless you changed the J definition

wrong sign

$[J_{\alpha\beta}, J_{\rho\sigma}]^\mu{}_\nu = i(g_{\sigma\beta}J_{\alpha\rho} - g_{\rho\alpha}J_{\sigma\beta} - g_{\beta\sigma}J_{\alpha\sigma} + g_{\alpha\sigma}J_{\rho\beta})$

two equal index

with $J_{\alpha\beta} = -J_{\beta\alpha}$

again

you picked up an extra minus sign here

⇒ $\boxed{i[J_{\alpha\beta}, J_{\rho\sigma}] = (g_{\sigma\beta}J_{\alpha\beta} + g_{\rho\alpha}J_{\beta\sigma} + g_{\beta\sigma}J_{\sigma\alpha} + g_{\alpha\sigma}J_{\rho\beta})}$

(b) from $i[J_{\alpha\beta}, J_{\gamma\sigma}] = (g_{\sigma\beta} J_{\alpha\gamma} + g_{\gamma\alpha} J_{\beta\sigma} + g_{\beta\sigma} J_{\gamma\alpha} + g_{\alpha\sigma} J_{\gamma\beta})$
again
and as discussed in class, the generators of the Lorentz group comprehend rotation and boost

So, it's possible to split $J_{\mu\nu}$ into:

rotations : $J_i = 1/2\, \epsilon_{ijk} J_{jk}$
boosts : $K_i = J_{0i}$

so with $[J_i, J_j] = \frac{1}{4} \epsilon_{ike} \epsilon_{jmn} [J_{ke}, J_{mn}]$

missing an i: $[J,J] \propto iJ$

$= 1/4\, \epsilon_{ike} \epsilon_{jmn} (g_{ne} J_{ke} + g_{mk} J_{en} + g_{en} J_{nk} + g_{kn} J_{me})$

$= i \epsilon_{ijk} J_k$ how? Should have detailed better the steps.

doing the same for the combinations

this derivation was part of the exercise.

$$\begin{cases} [J_i, J_j] = i\epsilon_{ijk} J_k \\ [J_i, K_j] = i\epsilon_{ijk} K_k \\ [K_i, K_j] = -i\epsilon_{ijk} J_k \end{cases} \quad \text{(VIII)}$$

from this the Lorentz algebra can be written in terms of both generators where

$$\boxed{A_i = \frac{1}{2}(J_i + iK_i)} \quad , \quad \boxed{B_i = \frac{1}{2}(J_i - iK_i)}$$

and using (VIII) we get to

$$\begin{cases} [A_i, A_j] = i\epsilon_{ijk} A_k \\ [B_i, B_j] = i\epsilon_{ijk} B_k \\ [A_i, B_j] = 0 \end{cases}$$

so here, the representation is irreducible

it was also necessary to derive these expressions.

As well to argue that A, B are two irreducible su(2) algebras, and thus the irrep are given by a pair of half integers

For the Nöther's Theorem exercises

As known

$$\delta S = S' - S = \int d^4x'\, \mathcal{L}'(\phi'(x'), \partial_\mu(\phi'(x'))) - \int d^4x\, \mathcal{L}(\phi(x), \partial_\mu(\phi(x)) = 0$$

with $x'^\mu = x^\mu + \delta x$ and $d^4x' = \left|\det\left(\frac{\partial x'^\mu}{\partial x^\nu}\right)\right| d^4x$

I

$$\frac{\partial x'^\mu}{\partial x^\nu} = \frac{\partial}{\partial x^\nu}(x^\mu + \delta x) = \delta^\mu_\nu + \partial_\nu \delta x^\mu \Rightarrow \boxed{d^4x' = d^4x(1 + \partial_\mu \delta x^\mu)}$$

ok!

with this

$$\Rightarrow \delta S = \int d^4x \left[\mathcal{L}'(\phi'(x'), \partial_\mu(\phi'(x')))(1 + \partial_\mu \delta x^\mu) - \mathcal{L}(\phi(x), \partial_\mu(\phi(x))\right]$$

defining
$$\mathcal{L}'(x) \equiv \mathcal{L}'(\phi'(x), \partial_\mu(\phi'(x)))$$
$$\mathcal{L}'(x') \equiv \mathcal{L}'(\phi'(x'), \partial_\mu(\phi'(x')))$$
to reduce notation

$$\mathcal{L}'(x') = \mathcal{L}'(x + \delta x) \overset{\text{Taylor expanding}}{\simeq} \mathcal{L}'(x) + (\partial_\mu \mathcal{L})\delta x^\mu$$

$$\Rightarrow \delta S = \int d^4x \left[(\mathcal{L}'(x) + (\partial_\mu \mathcal{L})\delta x^\mu)(1 + \partial_\mu \delta x^\mu) - \mathcal{L}(x)\right]$$

$$= \int d^4x \left[\underbrace{\mathcal{L}'(x) - \mathcal{L}(x)}_{\delta \mathcal{L}} + (\partial_\mu \mathcal{L})\delta x^\mu + \mathcal{L}(x)\partial_\mu \delta x^\mu + \cancel{O}(\delta x^2)\right]$$

Now expanding
$$\delta \mathcal{L} = \frac{\partial \mathcal{L}}{\partial \phi(x)}\delta \phi(x) + \frac{\partial \mathcal{L}}{\partial \partial_\nu \phi(x)}\overbrace{\delta(\partial_\nu \phi(x))}^{= \partial_\nu \delta \phi(x)}$$

$$\rightarrow \delta S = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi(x)}\delta \phi(x) + \frac{\partial \mathcal{L}}{\partial \partial_\nu \phi(x)}\partial_\nu(\delta \phi(x)) + \underbrace{(\partial_\mu \mathcal{L})\delta x^\mu + \mathcal{L}\,\partial_\mu \delta x^\mu}_{\partial_\mu(\mathcal{L}\,\delta x^\mu)}\right]$$

by product rule
$$\frac{\partial \mathcal{L}}{\partial \partial_\nu \phi(x)}\partial_\nu(\delta \phi(x)) = \partial_\mu\left(\frac{\partial \mathcal{L}\,\delta \phi(x)}{\partial(\partial_\mu \phi(x))}\right) - \left(\partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi(x))}\right)\delta \phi(x)$$

and with Euler - Lagrange equation $\partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi(x))} = \frac{\partial \mathcal{L}}{\partial \phi(x)}$

$$\Rightarrow \delta S = \int d^4x \left[\cancel{\frac{\partial \mathcal{L}}{\partial \phi(x)} \delta\phi(x)} + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi(x))} \delta\phi(x) \right) - \underbrace{\left(\partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi(x))} \right)}_{= \cancel{\frac{\partial \mathcal{L}}{\partial \phi(x)} \delta\phi(x)}} \delta\phi(x) + \partial_\mu (\mathcal{L}\, \delta x^\mu) \right]$$

$$\Rightarrow \delta S = \int d^4x \left[\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi(x))} \delta\phi(x) \right) + \partial_\mu (\mathcal{L}\, \delta x^\mu) \right] = 0$$

$$\Rightarrow \int d^4x\, \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi(x))} \delta\phi(x) + \mathcal{L}\, \delta x^\mu \right] = 0$$

So, according to the Nother theorem, under a symmetry the conserved current is going to be

$$\boxed{J^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi(x))} \delta\phi(x) + \mathcal{L}\, \delta x^\mu}$$

Perfect! Superb proof of Noether's theorem.

2 Consider the four vector electromagnetic potential $A^\mu = (\phi, \vec{A})$

1,5/2

with $\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$ and $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$

let's recover that

$$S[A^\mu] = \int dt \int d^3x\, \mathcal{L}(A^\mu, \partial A^\mu)$$

so, let's vary the action $\Rightarrow$ $\boxed{\delta S[A^\mu] = 0}$

$A^\mu \to A^\mu + \delta A^\mu$ and $\partial A^\mu \to \partial A^\mu + \partial \delta A^\mu$

$$\delta S = \int d^3x \int dt \left\{ \mathcal{L}(A^\mu + \delta A^\mu, \partial_\mu A^\mu + \partial_\mu \delta A^\mu) - \mathcal{L}(A^\mu, \partial A^\mu) \right.$$

$$= \int d^4x \left\{ \frac{\partial \mathcal{L}}{\partial A^\mu} \delta A^\mu + \boxed{\frac{\partial \mathcal{L}}{\partial \partial_\mu A^\mu} \partial \delta A^\mu} \right\} \cdot$$

using integral by parts.

$$\int_\Omega d^4x \underbrace{\frac{\partial \mathcal{L}}{\partial \partial_\mu A^\mu}}_{u} \underbrace{\partial \delta A^\mu}_{dv} = \int_{\partial\Omega} d^4x\, \partial_\mu \cancel{\left(\frac{\partial \mathcal{L}}{\partial \partial_\mu A^\mu} \delta A^\mu \right)}^{0} - \int_\Omega d^4x\, \delta A^\mu\, \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu A^\mu} \right)$$

$$= \int d^4x \left\{ \frac{\partial \mathcal{L}}{\partial A^\mu} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu A^\mu} \right) \right\} \delta A^\mu = 0 \Rightarrow$$

$$\Rightarrow \boxed{\frac{\partial \mathcal{L}}{\partial A^\mu} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu A^\mu} \right) = 0}$$

Now, with the Euler-Lagrange equation for the potential A^μ. Let's apply the Lagrangian on it.

(a) I, Term $\frac{\partial \mathcal{L}}{\partial A^\mu} = \frac{\partial}{\partial A^\mu} \left[-\frac{1}{4} F_{\rho\nu} F^{\rho\nu} \right]$

$$= -\frac{1}{4} \frac{\partial}{\partial A^\mu} \left[(\partial_\rho A_\nu - \partial_\nu A_\rho)(\partial^\rho A^\nu - \partial^\nu A^\rho) \right]$$

but any term $\frac{\partial}{\partial A^\mu}(\partial_\rho A_\nu) = 0 \Rightarrow \frac{\partial \mathcal{L}}{\partial A^\mu} = 0$

II, Term $\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu A_\nu} \right) = -\frac{1}{4} \partial_\mu \left[\frac{\partial}{\partial \partial_\mu A_\nu} (\partial_\rho A_\sigma - \partial_\sigma A_\rho)(\partial^\rho A^\sigma - \partial^\sigma A^\rho) \right]$

$$= -\frac{1}{4} \partial_\mu \left[\frac{\partial}{\partial \partial_\mu A_\nu} (\partial_\rho A_\sigma)(\partial^\rho A^\sigma) - (\partial_\rho A_\sigma)(\partial^\sigma A^\rho) - (\partial_\sigma A_\rho)(\partial^\sigma A^\rho) + (\partial_\sigma A_\rho)(\partial^\sigma A^\rho) \right]$$

$$= \frac{-1}{4}\partial_\mu\Big[\delta^\mu_\rho\delta^\nu_\gamma(\partial^\rho A^\gamma) + g^{\mu\rho}g^{\nu\gamma}(\partial_\rho A_\gamma) - \delta^\mu_\rho\delta^\nu_\gamma(\partial^\gamma A^\rho) - g^{\mu\gamma}g^{\nu\rho}(\partial_\rho A_\gamma)$$

$$- \delta^\mu_\rho\delta^\nu_\gamma(\partial^\gamma A^\rho) - g^{\mu\gamma}g^{\nu\rho}(\partial_\rho A_\gamma) + \delta^\mu_\gamma\delta^\nu_\rho(\partial^\gamma A^\rho) + g^{\mu\gamma}g^{\nu\rho}(\partial_\gamma A_\rho)\Big]$$

$$= -\frac{1}{4}\partial_\mu\left[\partial^\mu A^\nu + \partial^\mu A^\nu - \partial^\nu A^\mu - \partial^\nu A^\mu - \partial^\nu A^\mu - \partial^\nu A^\mu + \partial^\mu A^\nu + \partial^\mu A^\nu\right]$$

$$= -\frac{1}{4}\partial_\mu\left[4(\partial^\mu A^\nu - \partial^\nu A^\mu)\right] = -\partial_\mu(F^{\mu\nu}) \Rightarrow \boxed{\partial_\mu F^{\mu\nu} = 0}$$

ok!

(b) from $\delta S = \int d^4x \frac{d}{dx^\sigma}\left[\frac{\partial \mathcal{L}}{\partial(\partial_\sigma \phi_j)}\delta\phi_j(x) + \mathcal{L}\,\Delta x^\sigma\right]$

the conserved current is $J^{\mu} = \frac{\partial \mathcal{L}}{\partial(\partial^\nu A^\mu)}\delta A^\mu(x) + \mathcal{L}\,\delta x^\nu$

free μ free ν contracted μ

watch out for index placement

for a translation $\boxed{\delta x^\nu = x'^\nu - x^\nu = (x^\nu + a^\nu) - x^\nu = a^\nu}$

Beyond that, for translations, $A'^\mu(x') = A^\mu(x)$
and with $x' = x + \delta x$
$\rightarrow \boxed{A'^\mu(x') = A^\mu(x' - \delta x)}$ ‡

Now varing the field

$$\delta A^\mu(x) = A'^\mu(x) - A^\mu(x) = \underline{A^\mu(x - \delta x)} - A^\mu(x)$$

Taylor expanding it

$$= A^\mu(x) - \partial_\nu A^\mu(x)\,\delta x^\nu - A^\mu(x)$$

$$\Rightarrow \boxed{\delta A^\mu(x) = -\partial_\nu A^\mu(x)\,\delta x^\nu} = -\partial_\nu A^\mu(x)\,a^\nu$$

good

so replacing on the conserved current

$$J^\nu = \frac{\partial \mathcal{L}}{\partial(\partial_\nu A^\mu)} \delta A^\mu(x) + \mathcal{L}\, \delta x^\nu = -\frac{\partial \mathcal{L}}{\partial(\partial_\nu A^\mu)} \partial_\rho A^\mu(x)\, a^\rho + \mathcal{L}\, a^\nu$$

where $J^{\rho\nu}$ is the EM tensor!

$$J^\nu = -a^\rho \left[\frac{\partial \mathcal{L}}{\partial(\partial_\nu A^\mu(x))} \partial_\rho A^\mu(x) - \mathcal{L}\, \delta_\rho^\nu \right] \Rightarrow \boxed{J^{\rho\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\nu A^\mu(x))} \partial^\rho A^\mu(x) - \mathcal{L}\, g^{\rho\nu}}$$

usually is defined with $J^{\nu\rho}$

ok. But it was expected to explicitly compute it

③ Considering $\mathcal{L} = \frac{1}{2} \partial_\mu \phi\, \partial^\mu \phi - \frac{\lambda}{4!} \phi^4$

1.9/2

with
$$\begin{cases} x \to x' = e^\epsilon x \\ \phi(x) \to \phi'(x') = e^{-\epsilon} \phi(x) \end{cases}$$

According to the lectures, a symmetry is a transformation that takes solution of EOM to solution, therefore, leaves S unchanged.

so, we want to $\delta S = 0$

so $\delta S = \int d^4x'\, \mathcal{L}'(\phi'(x'), \partial_\mu \phi'(x')) - \int d^4x\, \mathcal{L}(\phi(x) - \partial_\mu \phi(x))$

because of $x' = e^\epsilon x \Rightarrow \partial'_\mu = \frac{\partial}{\partial x'^\mu} = \frac{\partial}{\partial(e^\epsilon x^\mu)} = e^{-\epsilon} \frac{\partial}{\partial x^\mu} = e^{-\epsilon} \partial_\mu$

$$\delta S = \int d^4x' \left(\frac{1}{2} \partial'_\mu \phi' \partial'^\mu \phi' - \frac{\lambda}{4!} \phi'^4 \right) - \int d^4x \left(\frac{1}{2} \partial_\mu \phi\, \partial^\mu \phi - \frac{\lambda}{4!} \phi^4 \right)$$

$$= \int d^4x' \left[\left(\frac{e^{-4\epsilon}}{2} \partial_\mu \phi\, \partial^\mu \phi - \frac{e^{-4\epsilon}}{4!} \lambda \phi^4 \right) - \int d^4x \left(\frac{1}{2} \partial_\mu \phi\, \partial^\mu \phi - \frac{\lambda}{4!} \phi^4 \right) \right.$$

and $d^4x' = \det\left(\frac{\partial x'^\mu}{\partial x^\nu} \right) d^4x = \det(e^\epsilon \mathbb{1})\, d^4x = e^{4\epsilon} d^4x$

$$= \int d^4x \left[e^{4\epsilon} \left(\frac{e^{-4\epsilon}}{2} \partial_\mu \phi\, \partial^\mu \phi - \frac{e^{-4\epsilon}}{4!} \lambda \phi^4 \right) - \left(\frac{1}{2} \partial_\mu \phi\, \partial^\mu \phi - \frac{\lambda}{4!} \phi^4 \right) \right]$$

$$= \int d^4x \left[\left(\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{\lambda}{4!} \phi^4 \right) - \left(\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{\lambda}{4!} \phi^4 \right) \right] = 0$$

So, this is a symmetry. OK

from this, finding the conserved current $J^\mu = \dfrac{\partial \mathcal{L}}{\partial(\partial^\nu \phi(x))} \delta \phi(x) + \mathcal{L} \delta x^\nu$

from $x^\mu \to x'^\mu = e^\epsilon x^\mu$, the infinitesimal transformation is

$$x'^\mu = x^\mu + \epsilon x^\mu$$

with this $\boxed{\delta x^\mu = x'^\mu - x^\mu = \epsilon x^\mu}$

and for $\phi(x) \to \phi'(x') = e^{-\epsilon} \phi(x)$

the infinitesimal transformation is

$$\phi'(x') = \phi(x) - \epsilon \phi(x)$$

$$\delta \phi = \phi'(x) - \phi(x) = \overbrace{\phi'(x' - \delta x)}^{\text{taylor expanding}} - \phi(x) = \phi'(x') - \partial_\mu \phi'(x') \delta x - \phi(x)$$

$$= \cancel{\phi(x)} - \epsilon \phi(x) - \partial_\mu (\phi(x) - \underbrace{\cancel{\epsilon}}_{\partial_\mu \epsilon \sim 0} \phi(x)) \delta x - \cancel{\phi(x)} = -\epsilon \phi(x) - \partial_\mu \phi(x) \epsilon x^\mu$$

wrong sign — ok

$$\Rightarrow \boxed{\delta \phi = -\epsilon (\phi(x) - \partial_\mu \phi(x) x^\mu)}$$

by this $J^\nu = \dfrac{\partial \mathcal{L}}{\partial(\partial^\nu \phi(x))} \delta \phi(x) + \mathcal{L} \delta x^\nu$

$$= \frac{\partial \mathcal{L}}{\partial(\partial^\nu \phi(x))} - \epsilon (\phi(x) - \partial_\mu \phi(x) x^\mu) + \mathcal{L} \epsilon x^\nu$$

from $\mathcal{L} = \dfrac{1}{2} \partial_\mu \phi \partial^\mu \phi - \dfrac{\lambda}{4!} \phi^4$

$$\frac{\partial \mathcal{L}}{\partial(\partial^\nu \phi(x))} = \frac{\partial}{\partial(\partial^\nu \phi(x))} \left(\frac{1}{2} \partial_\mu \phi \partial^\mu \phi \right) = \partial^\nu \phi$$

$$= \partial^\nu \phi \left[-\epsilon \left(\phi(x) - \partial_\mu \phi(x) x^\mu \right) + \mathcal{L} \epsilon x^\nu \right]$$

$$= \partial^\nu \phi \left[-\epsilon \left(\phi(x) - \partial_\mu \phi(x) x^\mu \right) \right] + \left(\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{\lambda}{4!} \phi^4 \right) x^\nu \epsilon$$

factor out $-\epsilon$

taking ϵ arbitrary

here is missing a minus sign

$$\boxed{J^\nu = \partial^\nu \phi \left(\phi(x) - \partial_\mu \phi x^\mu \right) + \left(\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{\lambda}{4!} \phi^4 \right) x^\nu}$$

4 Considering $x'^\mu = \Lambda^\mu{}_\nu x^\nu$ (I) $1{,}5/2$

with a infinitesimal transformation $x'^\mu = x^\mu + \delta\omega^\mu{}_\nu x^\nu$ (II)

a) Show that $\delta\omega_{\mu\nu} = -\delta\omega_{\nu\mu}$

for $x'^\mu = x^\mu + \delta\omega^\mu{}_\nu x^\nu = \Lambda^\mu{}_\nu x^\nu$

$\Rightarrow \delta^\mu{}_\nu x^\nu + \delta\omega^\mu{}_\nu x^\nu = \Lambda^\mu{}_\nu x^\nu \Rightarrow$

$\Rightarrow \quad \Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \underbrace{\delta\omega^\mu{}_\nu}_{<<1}$

now

$g_{\mu\nu} x'^\mu x'^\nu = g_{\mu\nu}\Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma x^\rho x^\sigma = g_{\rho\sigma} x^\rho x^\sigma \Rightarrow$

$\Rightarrow \quad g_{\rho\sigma} = g_{\mu\nu}\Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma$

$\cancel{g_{\rho\sigma}} = g_{\mu\nu}\Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma = g_{\mu\nu}(\delta^\mu{}_\rho + \delta\omega^\mu{}_\rho)(\delta^\nu{}_\sigma + \delta\omega^\nu{}_\sigma)$

$= g_{\mu\nu}\delta^\mu{}_\rho\delta^\nu{}_\sigma + g_{\mu\nu}\delta^\mu{}_\rho\delta\omega^\nu{}_\sigma + g_{\mu\nu}\delta\omega^\mu{}_\rho\delta^\nu{}_\sigma + O(\delta\omega^2)$

$= g_{\rho\sigma} + \delta^\mu{}_\rho\delta\omega_{\mu\sigma} + \delta^\nu{}_\sigma\delta\omega_{\nu\rho}$

$= \cancel{g_{\rho\sigma}} + \delta\omega_{\rho\sigma} + \delta\omega_{\sigma\rho}$

$\delta\omega_{\rho\sigma} + \delta\omega_{\sigma\rho} = 0 \Rightarrow \boxed{\delta\omega_{\rho\sigma} = -\delta\omega_{\rho\sigma}}$ ok

b) for a rotation, analizing $x'^\mu = x^\mu + \delta\omega^\mu{}_\nu x^\nu$

firsty, because of the anti-symmetry
$\delta\omega_{\mu\nu} = -\delta\omega_{\mu\nu} \Rightarrow \delta\omega_{\mu\mu} = 0$

furthermore

In the term $\delta\omega_{\mu\nu} x^\nu$, a spatial rotation should not change the temporal component of the four vector, only the spatial ones

so $x'^0 = x^0 + \underbrace{\delta\omega^0_{\ \nu}}\, x^\nu$

↳ must be zero

so because $\delta\omega_{\mu\mu} = 0$, $\delta\omega_{0\nu} = 0 \Rightarrow \delta\omega_{0i} = 0$

and because $\delta\omega_{\mu\nu} = -\delta\omega_{\mu\nu} \Rightarrow \delta\omega_{0i} = \delta\omega_{i0} = 0$

now, the remaining components are $\delta\omega_{ij}$, where $\delta\omega_{ii} = 0$ and $\delta\omega_{ij} = -\delta\omega_{ji}$

defining the components of $\delta\omega_{ij}$ as rotations by θ_a

$\delta\omega_{12} = -\delta\omega_{21} \equiv \theta_3$, $\delta\omega_{13} = -\delta\omega_{31} \equiv \theta_2$, $\delta\omega_{32} = -\delta\omega_{23} \equiv \theta_1$

this can be summarized with the levi-civita symbol, so

$$\delta\omega_{ij} = \epsilon_{ijk}\theta_k$$

In conclusion, for a rotation $\delta\omega_{\mu\nu} = \begin{cases} 0, & \text{if } \mu \text{ or } \nu = 0 \text{ or} \\ \epsilon_{ijk}\theta_k, & \text{if } \mu = i \text{ and } \nu = j \end{cases}$

ⓒ For a scalar field $\phi(x)$ obtain the conserved current associated to a Lorentz transformation

for the scalar field condition $\Rightarrow \phi'(x') = \phi(x)$ on space time transformations

Let's again look for the conserved current $J^\mu = \frac{\partial \mathcal{L}}{\partial(\partial^\nu \phi(x))} \delta \phi(x) + \mathcal{L}\, \delta x^\nu$

firstly with $x'^\mu = x^\mu + \delta \omega^\mu{}_\nu x^\nu$

$$\delta x^\mu = x'^\mu - x^\mu = x^\mu + \delta\omega^\mu{}_\nu x^\nu - x^\mu \Rightarrow \boxed{\delta x^\mu = \delta \omega^\mu{}_\nu x^\nu}$$

And for $\delta\phi = \phi'(x) - \phi(x)$

as $\phi'(x') = \phi(x) \rightarrow \phi'(x') = \phi(x' - \delta x)$

$\rightarrow \phi'(x) = \phi(x - \delta x)$

taylor expand

$$\Rightarrow \delta\phi = \underbrace{\phi(x - \delta x)}_{\text{taylor expand}} - \phi(x) = \cancel{\phi(x)} - \partial_\mu \phi(x)\, \delta x - \cancel{\phi(x)}$$

$$\Rightarrow \delta\phi = - \partial_\mu \phi(x)\, \delta x$$

$$\Rightarrow \boxed{\delta\phi = - \partial_\mu \phi(x)\, \delta\omega^\mu{}_\nu x^\nu}$$

Coming back to the conserved current

$$J^\nu = \frac{\partial \mathcal{L}}{\partial(\partial^\nu \phi(x))} \delta\phi(x) + \mathcal{L}\, \delta x^\nu$$

$$= \frac{\partial \mathcal{L}}{\partial(\partial^\nu \phi(x))} \left(- \partial_\mu \phi(x)\, \delta\omega^\mu{}_\gamma x^\gamma\right) + \mathcal{L}\, \delta\omega^\nu{}_\gamma x^\gamma$$

$$= \delta\omega^\mu{}_\gamma x^\gamma \left[- \frac{\partial \mathcal{L}}{\partial(\partial^\nu \phi(x))} \partial_\mu \phi(x) + \delta_\mu^\nu \mathcal{L} \right] \quad T^{\rho\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\nu A^\mu(x))} \partial^\rho A^\mu(x) - \mathcal{L} g^{\rho\nu}$$

$$= - \delta\omega_{\mu\gamma} x^\gamma \underbrace{\left[\frac{\partial \mathcal{L}}{\partial(\partial^\nu \phi(x))} \partial^\mu \phi(x) + g^{\mu\nu} \mathcal{L} \right]}_{T^{\mu\nu}}$$

wrong sign

$T^{\mu\nu} \rightarrow$ energy momentum tensor as exercise 2

with this $\boxed{J^\nu = - \delta\omega_{\mu\gamma} x^\gamma T^{\mu\nu}}$ is the conserved current!

the current can't depend on $\delta\omega$. It is necessary to remove it, as it's skew-symmetric this indices: $J^{\nu\mu\gamma} = - x^\gamma T^{\mu\nu} + x^\mu T^{\gamma\nu}$

5 $1/2$

Knowing that $A'^{\mu}(x') = \Lambda^{\mu}{}_{\nu} A^{\nu}(x)$

based on the Lagrangian $\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$

where $F_{\mu\nu} = \partial_{\mu} A_{\nu} - \partial_{\nu} A_{\mu}$

Again using the infinitesimal Lorentz transformation

$$\Lambda^{\mu}{}_{\nu} = \delta^{\mu}_{\nu} + \delta w^{\mu}{}_{\nu}$$

Again by Noether's theorem the conserved current is

$$J^{\mu} = \frac{\partial \mathcal{L}}{\partial(\partial^{\nu} A^{\mu})} \delta A^{\mu}(x) + \mathcal{L}\, \delta x^{\nu}$$

For the coordinates the Lorentz transformation is still valid

where $x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu}$, so, in the infinitesimal form $x'^{\mu} = x^{\mu} + \delta w^{\mu}{}_{\nu} x^{\nu}$

with this $\delta x^{\mu} = x'^{\mu} - x^{\mu} = \cancel{x^{\mu}} + \delta w^{\mu}_{\nu} x^{\nu} - \cancel{x^{\mu}} \Rightarrow \boxed{\delta x^{\mu} = \delta w^{\mu}{}_{\nu} x^{\nu}}$

and for the fields $\delta A^{\mu}(x) = A'^{\mu}(x) - A^{\mu}(x)$

with $x = x' - \delta x$

$$\delta A^{\mu}(x) = \underbrace{A'^{\mu}(x' - \delta x)}_{\text{taylor expanding}} - A^{\mu}(x)$$

wrong

taylor exp. $A^{\mu}(x') = A^{\mu}(x) + \delta x^{\alpha} \partial_{\alpha} A^{\mu}(x)$

$$= A'^{\mu}(x') - \partial_{\rho} A'^{\rho}(x') \delta x^{\mu} - A^{\mu}(x)$$

$\partial_{\mu} \delta w^{\mu}{}_{\nu} \sim 0$

$$= (\delta^{\mu}{}_{\nu} + \delta w^{\mu}{}_{\nu}) A^{\nu}(x) - \partial_{\rho} (\delta^{\rho}_{\sigma} + \delta w^{\rho}_{\sigma}) A^{\sigma}(x) \delta x - A^{\mu}(x)$$

$$= \delta w^{\mu}{}_{\nu} A^{\nu}(x) - \partial_{\sigma} A^{\sigma}(x) \delta x^{\mu}$$

$$= \delta w^{\mu}{}_{\nu} A^{\nu}(x) - \partial_{\sigma} A^{\sigma}(x) \delta w^{\mu}_{\rho} x^{\rho}$$

$$\Rightarrow \boxed{\delta A^{\mu}(x) = \delta w^{\mu}{}_{\nu} A^{\nu}(x) - \partial_{\sigma} A^{\sigma}(x) \delta w^{\mu}{}_{\rho} x^{\rho}}$$

now, going back to the conserved current.

$$J^\nu = \frac{\partial \mathcal{L}}{\partial(\partial_\nu A^\mu)} \delta A^\mu(x) + \mathcal{L}\, \delta x^\nu$$

should be $\partial_\rho A^\mu \delta\omega^\rho{}_\sigma x^\sigma$

$$= \frac{\partial \mathcal{L}}{\partial(\partial_\nu A^\mu)} [\delta\omega^\mu{}_\alpha A^\alpha(x) - \underbrace{\partial_6 A^6(x)\, \delta\omega^\mu{}_\rho x^\rho}] + \mathcal{L}(\delta\omega^\nu{}_\gamma x^\gamma)$$

as shown in ① a II $\frac{\partial \mathcal{L}}{\partial \partial_\mu A_\nu} = -F^{\mu\nu}$

$$\text{so } J^\nu = -F^\nu{}_\mu [\delta\omega^\mu{}_\alpha A^\alpha(x) - \partial_6 A^6(x)\, \delta\omega^\mu{}_\rho x^\rho] - \frac{1}{4} F_{\delta\theta} F^{\delta\theta} \delta\omega^\nu{}_\beta x^\beta$$

$$= [-\delta\omega^\mu{}_\alpha F^\nu{}_\mu A^\alpha(x) + F^\nu{}_\mu \partial_6 A^6(x)\, \delta\omega^\mu{}_\rho x^\rho] - \frac{1}{4} F_{\delta\theta} F^{\delta\theta} \delta\omega^\nu{}_\beta x^\beta$$

$$= -\omega^\mu{}_\rho \left[\delta^\rho{}_\alpha F^\nu{}_\mu A^\alpha(x) - F^\nu{}_\mu \partial_6 A^6(x)\, x^\rho + \frac{1}{4} F_{\delta\theta} F^{\delta\theta} \delta_\mu{}^\nu \delta^\rho_\beta x^\beta \right]$$

with $\omega^\mu{}_\rho$ arbitrary my conserved current is

$$\boxed{J^{\nu\mu}{}_\rho = \delta^\rho{}_\alpha F^\nu{}_\mu A^\alpha(x) - F^\nu{}_\mu \partial_6 A^6(x)\, x^\rho + \frac{1}{4} F_{\delta\theta} F^{\delta\theta} \delta_\mu{}^\nu \delta^\rho_\beta x^\beta}$$

as well as the error due to the taylor exp. There is also other mistake when factoring $\delta\omega$. It is skew-symmetric, thus, you must anti-symmetrize the entire expression to remove it. As example, pick the first term of your current:

$$J^\mu = -\omega_{\alpha\beta} F^{\mu\alpha} A^\beta = -\frac{\omega_{\alpha\beta}}{2} \underbrace{\left(F^{\mu\alpha} A^\beta - F^{\mu\beta} A^\alpha \right)}_{M^{\mu\alpha\beta}}$$